

# Module 4

## Circuit Analysis in the S Domain

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# Tutorial

1. Find the Laplace transform of each of the following functions:

$$(a) t^2 \cos(2t + 30^\circ) u(t) \quad (b) 3t^4 e^{-2t} u(t)$$

$$(c) 2tu(t) - 4 \frac{d}{dt} \delta(t) \quad (d) 2e^{-(t-1)} u(t)$$

$$(e) 5u(t/2) \quad (f) 6e^{-t/3} u(t)$$

$$(g) \frac{d^n}{dt^n} \delta(t)$$

2. Find the Laplace transform of the following signals:

$$(a) f(t) = (2t + 4)u(t)$$

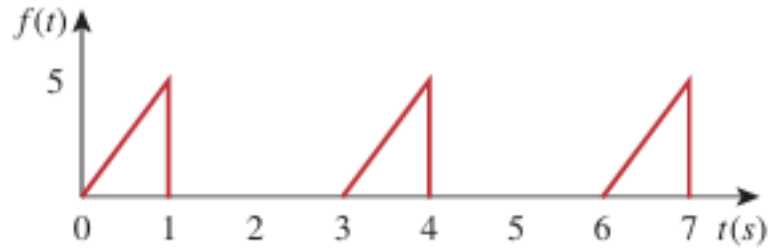
$$(b) g(t) = (4 + 3e^{-2t})u(t)$$

$$(c) h(t) = (6 \sin(3t) + 8 \cos(3t))u(t)$$

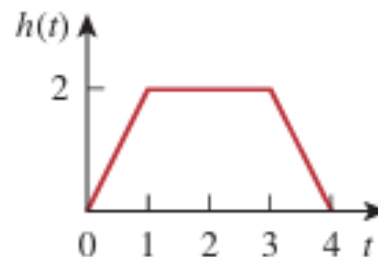
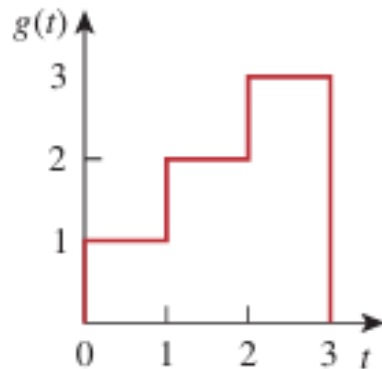
$$(d) x(t) = (e^{-2t} \cosh(4t))u(t)$$

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3. Determine the Laplace transform of the function in Figure shown below.



4. Obtain the Laplace transforms of the functions in the figures below.



# Tutorial

5. Determine the inverse Laplace transform of each of the following functions:

$$(a) \frac{8(s+1)(s+3)}{s(s+2)(s+4)}$$

$$(b) \frac{s^2 - 2s + 4}{(s+1)(s+2)^2}$$

$$(c) \frac{s^2 + 1}{(s+3)(s^2 + 4s + 5)}$$